METHOD 04 / A COMPLETE BEGINNER GUIDE

Robust clipped correlation

Question this method answers: How closely do two return series move together if we cap each day's return at ±c before correlating — so one crash day cannot dominate?

You still keep raw Pearson r for comparison. Clipping is a sensitivity check, not a replacement for viewing the full data.

Page map

Pages	Topic
2	Idea, sign, size, percent trap
3	FX return math (c in decimal units)
4	Which series clipped correlation uses
5	Research notes (section 04)
6	Symbol dictionary
7-8	Clip formula and hand example
9-10	Daily majors workflow and limits
11	Software reproduction
12-15	Sensitivity, practice, sources

Prepared for: 04 - Robust Clipped Correlation. README status: not started.

02 / THE BASIC IDEA

What is rho_clip?

Winsorize (clip) each return to a band, then compute Pearson correlation on the clipped series. Outliers remain in the sample but their *magnitude* is capped.

Sign and size

rho_clip uses the same -1 to +1 scale as Pearson. Sign still means direction of co-movement among clipped values.

Is this number a percent?

Mistake	Truth
"c = 0.02 means clip at 2% returns"	Yes in decimal return units: ±0.02 ≈ ±2% simple move size.
"rho_clip = 0.5 means half the days"	No. It is still a correlation, not a day fraction.

FX percent math (majors)

```
Simple return = P_t / P_{t-1} - 1
Log return = ln(P_t / P_{t-1})
```

Clip c is applied in the same units as the return series you feed in. If returns are decimals, c=0.02 caps at about ±2%. Example price move 1.1000→1.1110 is +1% — inside the band.

Quote reversal flips log signs. Missing ≠ 0. Four closes → three returns.

04 / DATA CHOICE

Which series this method uses

Use **matched daily returns** (log or simple — pick one and document it). Clip each leg separately: x_tilde and y_tilde, then correlate.

Always report raw Pearson r on unclipped returns alongside rho_clip so readers see what the cap changed.

If you clip log returns, interpret c on the log scale (not identical to simple-percent caps). Stay consistent across both symbols and across time when comparing experiments.

05 / CONSTRUCTION IN THE REPO SHEET

Research notes (section 04)

```
x_tilde = min(c, max(-c, x))
y_tilde = min(c, max(-c, y))
rho_clip = Corr(x_tilde, y_tilde)
```

Section 04 in forex-formulas-explained.pdf describes this construction. Pearson on clipped data matches scipy.stats.pearsonr applied to the clipped arrays.

Clipping discards information about crash *size* — two days both beyond +c look identical. That is intentional for robustness, but it can hide risk.

06 / SMALL MATH DICTIONARY

Symbol dictionary

Symbol	Meaning
x, y	Raw daily returns.
c	Clip bound (positive).
x_tilde, y_tilde	Clipped returns.
rho_clip	Pearson r on clipped series.
r	Raw Pearson on unclipped series.
n	Count of matched days in the window.
cap-hit count	Days where |return| > c before clipping — report separately.

Clipping is symmetric: positive and negative tails use the same c. After clipping, means and standard deviations change, so rho_clip can move even when rank order stays similar.

07 / CLIP THEN CORRELATE

Formula step by step

```
x_tilde_i = min(c, max(-c, x_i))
y_tilde_i = min(c, max(-c, y_i))
rho_clip = Pearson(x_tilde, y_tilde)
```

Clipping is applied row-wise before any mean or sum. Constants (all values inside the band) make correlation undefined — same rule as Pearson.

The min/max form never drops rows: a -5% day stays in the sample as -c. That differs from deleting outliers, which would change n.

Compute raw r on the same rows for fair comparison; do not clip one leg only unless that asymmetry is part of the research design.

08 / C = 0.02

Hand example (teaching)

Small made-up decimal returns:

Day	x	x_tilde	y	y_tilde
1	0.01	0.01	0.008	0.008
2	-0.05	-0.02	-0.04	-0.02
3	0.003	0.003	0.001	0.001
4	0.02	0.02	0.015	0.015

```
rho_clip (computed) ≈ 0.9986
(raw x had a -5% day clipped to -2%)
```

The crash day still counts, but its push on r is limited. Compare to raw r on x, y to quantify sensitivity.

09 / HOW TO USE IT

Daily majors workflow

1. Choose c from economic sense

Try c = 0.02 (±2%) as a starting sensitivity line for daily majors, then sweep c = 0.01 and 0.03 in experiments.

2. Pair with raw Pearson

If raw r is high but rho_clip collapses, outliers or crisis days may be driving the story.

3. Document both

Future experiments under experiments/v001/ should store r, rho_clip, c, and n in NOTES.

Limits

rho_clip is not a VaR model, not a hedge ratio, and not proof of stability. It hides tail sizes — two different crises can look the same after clipping.

Choosing c after seeing results can overfit. Fix c before looking at pair outcomes when possible.

If almost every day hits the cap, rho_clip reflects mainly sign agreement inside a flat band — read cap-hit counts before interpreting the score.

Software check

```
c = 0.02
x = [0.01, -0.05, 0.003, 0.02]
y = [0.008, -0.04, 0.001, 0.015]
x_t = [min(c, max(-c, v)) for v in x]
y_t = [min(c, max(-c, v)) for v in y]
from scipy.stats import pearsonr
pearsonr(x_t, y_t)
```

Built-in assert target rho_clip = 0.998577 for the teaching table.

12 / WHY KEEP RAW R

Sensitivity mindset

Report: date range, n, raw r, rho_clip at stated c, and how many days hit the cap on each leg. Days hitting the cap are flags for manual review.

Run a small grid (c = 0.01, 0.02, 0.03) in experiments to see whether conclusions are stable. If rho_clip jumps when c moves slightly, the pair may be outlier-driven.

Document the grid in NOTES rather than cherry-picking the c that makes a pair look best.

13 / QUICK CHECKS

Practice

Q: x = 0.10 with c = 0.02 — x_tilde? **A:** 0.02.

Q: Does clipping add data? **A:** No, it transforms existing rows.

Q: x = -0.03 with c = 0.02 — x_tilde? **A:** -0.02.

Q: If raw r = 0.9 and rho_clip = 0.4, what might that mean? **A:** A few extreme days may be driving raw Pearson — inspect cap hits.

Always clip both legs with the same c unless you document an asymmetric design.

Sources

Links checked when this guide was built. Bracket numbers point to the references below. Teaching examples and cap-hit logic in this PDF are original walkthrough material for method 04.

[1] SciPy documentation: scipy.stats.pearsonr.
Used on clipped arrays for rho_clip.
Open source

[2] forex-formulas-explained.pdf — section 04 (this repository).
Clipped correlation construction.
Open source

Reading tip: confirm c in decimal return units, then reproduce rho_clip on the teaching table before running full major-pair panels.

Folder checklist

Freeze c, return type, and cap-hit counts. Keep raw and clipped correlations in the same table.

When experiments/v001/ lands, mirror method 01 pair lists so clipped and raw columns line up row-for-row on date.

Rebuild this PDF after formula edits: python build_guide.py from this folder.

End of robust clipped correlation beginner guide.